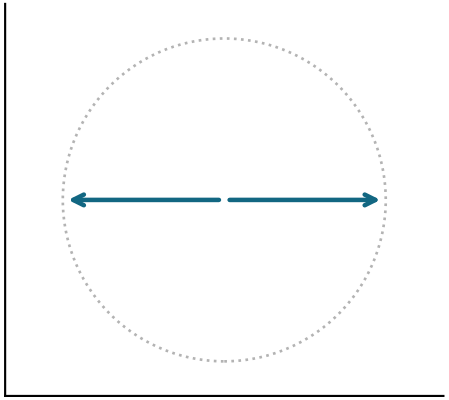
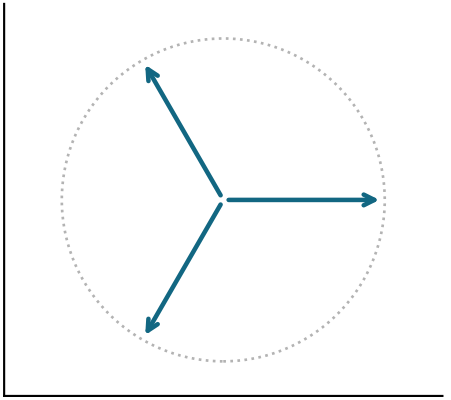
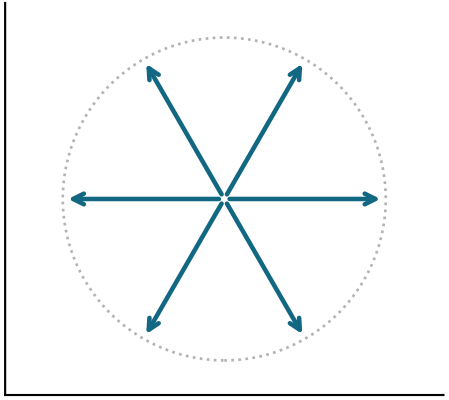
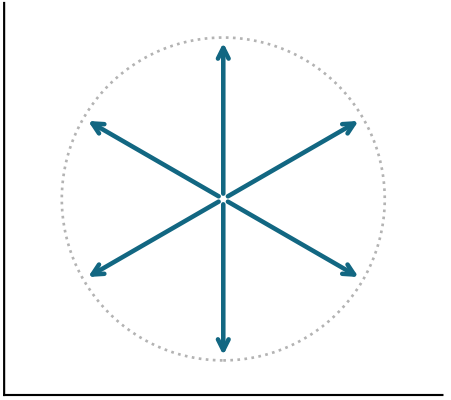
I: M (old X), two waves



II: K (old J), three waves



III: Gamma, first nonzero shell



IV: Gamma, next shell